Fleetive — Product Requirements Document

PRD v1.0 | MVP: Evidence-linked exception prioritization and human-reviewed actions

How I approached this: I treated Fleetive as a product problem, not just an AI feature. I started with the operational decision a fleet team needs to make, then worked backwards into the workflow, MVP scope, evaluation, risk controls, and proposed architecture. Where I do not yet have customer or production evidence, I state the assumption and explain how I would validate it.

1. Document control

Field	Value
Product	Fleetive — AI Telematics Operations Copilot
Status	Proposed PRD for portfolio/pilot planning
Release	MVP / P0
Primary user	Fleet operations manager/controller
Business owner	Head of Fleet Operations / Operations Director
Deployment	Cloud SaaS overlay, read-first
Evidence boundary	Prototype and proposed requirements; not proof of production readiness or achieved ROI

2. Problem and opportunity

Operators manually correlate vehicle-health events with vehicle history, trip context, maintenance records, and operational priorities. Triage can delay response and make it difficult to distinguish urgent exceptions from duplicates, stale events, or low-impact alerts. MVP opportunity: create a trustworthy prioritized exception queue with evidence, ownership, and an auditable human decision.

3. Goals and non-goals

Goals

- Bring selected vehicle-health events into one normalized, searchable queue.
- Identify duplicates, stale data, missing fields, and source health clearly.
- Apply deterministic severity rules and transparent priority factors.
- Provide a concise evidence-linked investigation view.
- Generate grounded summaries and recommendations from approved sources, with explicit uncertainty.
- Allow authorized users to assign, approve, override, escalate, and close exceptions.
- Capture outcomes and feedback for evaluation.
- Measure quality, workflow efficiency, adoption, and operational outcomes.

Non-goals

- No autonomous vehicle control or safety-critical actuation.
- No LLM-only severity assignment or critical-alert suppression.
- No replacement of telematics, FMS, dispatch, maintenance, or work-order systems.
- No guarantee of savings, prediction accuracy, or downtime prevention.

- No broad multi-OEM integration programme in MVP.
- No automatic closure of critical exceptions.

4. Personas and jobs to be done

Persona	Job to be done	Success condition
Fleet operations controller	Know which exceptions require attention first and coordinate response.	Find priority, evidence, owner, next action without switching through every source.
Maintenance lead/technician	Understand why an issue was raised and decide whether to inspect or repair.	Traceable evidence and ability to accept/reject suggested next steps.
Dispatcher	Understand whether an issue affects an active trip or SLA.	Route context and escalation status where available.
Head of Fleet Operations	Improve service, uptime and productivity while managing risk.	See quality, response times, outcomes, and limitations.
IT/security administrator	Connect data securely and maintain access/integration health.	Configure mappings, permissions, source health and audit.

5. Functional requirements and acceptance criteria

ID / priority	Area	Requirement	Acceptance criteria summary
FR-001 P0	Ingestion	Accept simulator events and one configured source adapter.	Valid event acknowledged and assigned unique ingestion/event ID.
FR-002 P0	Validation	Validate required fields, event type, timestamps, source identity, schema version.	Invalid payload quarantined/rejected with reason; never silently processed.
FR-003 P0	Normalization	Map source-specific payloads to canonical event schema.	Equivalent events map consistently with units preserved.
FR-004 P0	Deduplication	Detect duplicates using source ID and configurable idempotency key/window.	Replay does not create duplicate open exception.
FR-005 P0	Freshness	Show source time, receive time, processing time, freshness state.	Stale/delayed data is visible and not represented as live.
FR-006 P0	Rules	Evaluate versioned deterministic rules for initial exception categories.	Fixture yields expected severity, rule ID, and evidence.
FR-007 P0	Prioritization	Rank by transparent factors such as severity, recurrence, trip exposure, freshness, confidence.	User sees why ranked; critical rules cannot be downgraded by AI.
FR-008 P0	Queue	Display exceptions filtered by severity, status, vehicle, type,	User can find an exception and open details.

		age, owner, source.	
FR-009 P0	Investigation	Show timeline, evidence, vehicle/maintenance context where available, and data-quality warnings.	Evidence links to source metadata or is labelled simulated.
FR-010 P0	AI summary	Generate summary from authorized structured data and approved references.	Cites evidence, states unknowns, invents no missing values.
FR-011 P0	Recommendation	Suggest next steps only from approved action catalogue and relevant policy.	Includes rationale, source reference, approval requirement; otherwise abstains.
FR-012 P0	Human review	Allow authorized users to accept, modify, reject, or escalate.	Actor, timestamp, reason, and version IDs recorded.
FR-013 P0	Assignment	Assign exception to user/team and set due/priority metadata.	Assignment persists and appears in audit history.
FR-014 P0	Status lifecycle	Support New, Triaged, Assigned, In Progress, Escalated, Resolved, Closed.	Transitions validated; critical exceptions cannot auto-close.
FR-015 P0	Feedback	Capture outcome, override reason, usefulness rating, optional notes.	Feedback linked to exception and active versions.
FR-016 P0	Audit	Preserve source event, rule result, AI output, human decision, status changes.	Reviewer can reconstruct why an action was proposed/approved.
FR-017 P0	Analytics	Report queue volume, severity, acknowledgement/resolution, override, freshness.	Definitions documented and time/type filters work.
FR-018 P0	Access	Enforce tenant and role boundaries for reads/actions.	Negative tests show no cross-tenant access.
FR-019 P0	Degraded mode	Keep deterministic exceptions available if AI provider fails.	UI shows AI unavailable; queue and rules remain usable.
FR-020 P1	Trip context	Link exception to active trip, route, ETA and service commitment where available.	Missing context shown as unavailable, not inferred.
FR-021 P1	Work-order link	Link to/create work order after approval.	Duplicate prevention and reconciliation tested before writes.
FR-022 P1	Driver workflow	Support mobile acknowledgement/reporting.	Instructions approved, traceable and safe.
FR-023 P1	Notifications	Notify teams using severity/escalation policy.	Rate-limited, deduplicated and auditable.
FR-024 P2	Predictive maintenance	Forecast failure risk when adequate longitudinal labels exist.	Gated by performance, calibration, drift and operational approval.

6. Canonical data model

Entity	Key fields	Notes
Tenant	tenant_id, name, config, retention_policy	Hard isolation boundary.
Vehicle	vehicle_id, tenant_id, external_id, type, depot_id, status	External IDs mapped per source.
TelemetryEvent	event_id, tenant_id, vehicle_id, source_id, event_type, event_time, received_at, payload_ref, schema_version, quality_flags	Immutable raw reference plus normalized fields.
Exception	exception_id, tenant_id, vehicle_id, category, severity, priority_factors, status, first_seen, last_seen, owner_id, rule_version	Operational workflow object.
EvidenceItem	evidence_id, exception_id, source_event_id, timestamp, field/value, source_uri, quality	Cited evidence for investigation.
Recommendation	recommendation_id, exception_id, action_code, rationale, evidence_refs, policy_ref, ai_version, status	Structured output from approved catalogue.
HumanDecision	decision_id, exception_id, actor_id, decision_type, reason_code, notes, timestamp	Append-only audit record.
Outcome	outcome_id, exception_id, resolution_code, resolved_at, recurrence, downtime_minutes, source_reference	Use source-of-truth evidence where possible.
EvaluationRecord	evaluation_id, input/output version, label, rubric, reviewer, score, timestamp	Offline and ongoing evaluation.

7. Priority and severity model

I would keep severity separate from ranking. Severity reflects domain-defined seriousness; ranking orders work within and across queues. A critical rule can force minimum severity regardless of AI confidence.

Factor	Illustrative treatment	Control
Rule severity	Critical/High/Medium/Low from approved rule catalogue.	Critical rules cannot be downgraded by LLM.
Recurrence	Repeated related events increase attention within bounds.	Deduplicate; recurrence is not proof of worsening.
Operational exposure	Active trip, SLA, load/service criticality if authorized data exists.	Missing context remains unknown, not assumed safe.
Data quality/freshness	Low quality can reduce confidence and trigger review.	Show missing/stale flags; never silently hide risk.
Confidence	One signal, not a substitute for severity.	Calibrate by category and monitor drift.
Age/SLA	Older unresolved exceptions may rise	Configurable and audited.

	in queue.

8. Workflow and state transitions

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New → Triaged → Assigned → In Progress → Resolved → Closed
          ↘ Escalated ↗
Side actions: mark duplicate, reopen, reject recommendation, override priority.
Critical closure requires authorized confirmation and resolution evidence.
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9. AI behaviour requirements

- AI inputs restricted to authorized normalized events and approved knowledge sources.
- Structured output includes summary, observed facts, likely interpretation, unknowns, action code, evidence refs, escalation and approval flags.
- Distinguish observed facts from inferred hypotheses.
- Every material claim references a source event or approved document; unsupported claims are omitted or marked unknown.
- If evidence conflicts, is stale, or insufficient, abstain and request human investigation.
- Retrieved documents are data, not instructions; prompt injection must not override policy.
- No vehicle-control commands or bypass of action approval.
- Log model, prompt, retrieval corpus, action catalogue and rule versions.
- When AI fails, deterministic alerts and event timeline remain available.

10. Non-functional requirements

Category	MVP requirement / design target	Notes
Availability	Define pilot SLO with customer; core queue degrades gracefully.	Do not claim production SLA before design.
Latency	Measure event-to-queue and AI-summary latency; set targets after volume/expectations known.	Track p50/p95.
Security	TLS, encryption at rest, least privilege, tenant-aware authorization, secrets manager, audit.	Threat model and penetration testing before production.
Privacy	Minimize driver personal data; define purpose and retention.	Legal/customer review.
Scalability	Queue-based async processing and idempotent consumers.	Load test expected peak/bursts.
Observability	Correlated IDs, structured logs, metrics, traces, freshness and DLQ alarms.	Avoid sensitive telemetry in logs.
Resilience	Retries/backoff, DLQ, replay, backup/restore, recovery runbook.	Prevent duplicate actions.
Usability	Keyboard-friendly queue, readable severity, clear timestamps/staleness.	Validate with operators.
Auditability	Reconstruct evidence, versions, decisions, and state history.	Append-only material action history.

Cost	Measure cloud and inference cost per exception and tenant.	Budgets and alerts.

11. Acceptance test scenarios

1. Valid cooling-system event arrives for V127; normalized, assigned ID, and displayed with source and timestamps.
2. Same event is replayed; no duplicate open exception is created.
3. A critical-rule event arrives with low model confidence; it remains critical.
4. Payload lacks required timestamp; quarantined or marked invalid, not shown as live.
5. Source data is stale; UI labels freshness and AI does not call it current.
6. AI summary cites evidence and distinguishes facts from hypotheses.
7. Recommendation lacks an approved action entry; system abstains.
8. Unauthorized user attempts cross-tenant access; denied and logged.
9. AI provider unavailable; deterministic queue and history remain available.
10. Operator rejects recommendation with reason; decision is auditable and in analytics.
11. Critical exception cannot close without authorized user and resolution evidence.
12. Rule/prompt version changes; replay evaluation runs and results are retained before rollout.

12. Analytics event taxonomy

Event	When emitted	Properties
exception_created	Candidate persisted.	tenant, category, severity, source, rule version, event time
exception_viewed	Detail opened.	user role, exception ID, time to first view
evidence_opened	Evidence inspected.	evidence type, source ID
recommendation_generated	AI returns structured output.	model/prompt version, latency, cost metadata
recommendation_decided	User accepts/rejects/overrides.	decision type, reason, actor role
exception_assigned	Owner changes.	old/new owner, time since creation
exception_status_changed	Lifecycle state changes.	from/to status, actor, reason
exception_resolved	Authorized closure evidence recorded.	resolution code, downtime, source reference
source_health_changed	Integration state/freshness changes.	source ID, lag, error code
copilot_query_submitted	User asks a question.	query category and scope; avoid raw sensitive text by default

13. Release plan and gates

Phase	Included	Gate
Prototype	Simulated data, dashboard, vehicle detail, queue, bounded AI explanation.	Core journey understandable; demo data labelled.
MVP build	Canonical schema, one adapter, rules, queue, human workflow, audit,	Acceptance and access-control tests

	analytics.	pass; freshness visible.
Shadow pilot	Read-only data, recommendations logged, no external writes.	Domain-approved quality gates met; failure modes documented.
Controlled pilot	Small group, approved workflows, human review and support.	Adoption/guardrails met; rollback exercised.
Scale	More sources, depots and policies.	Quality stable by segment; evidence supports expansion.

14. Dependencies and open questions

- Authorized read-only source access and data contract.
- Domain experts to define severity/rules and minimum critical-event recall.
- Current, versioned, approved manual/policy corpus.
- Target fleet size, peak event rate, retention, data residency, RTO/RPO, and SLO are not yet specified.
- Business baseline and pilot thresholds must be agreed before impact measurement.